Missing Values

Our dataset

List of transactions in an on-line stores selling t-shirts

No of Items: **Number of items sold**

Price: **price per item**

Category: **t-shirt category**

	Date	No of Items	Price	Category
0	18.11.2021	3.0	10.0	NaN
1	19.11.2021	NaN	15.0	White M
2	NaN	NaN	10.0	White M
3	21.11.2021	NaN	10.0	Hoodie
4	22.11.2021	10.0	5.0	Black S
5	23.11.2021	NaN	20.0	White M
6	24.11.2021	NaN	NaN	Hoodie
7	25.11.2021	12.0	15.0	Black S
8	26.11.2021	NaN	NaN	Black S
9	27.11.2021	NaN	15.0	Black S
10	NaN	NaN	NaN	Hoodie
11	29.11.2021	NaN	10.0	Hoodie

Missing Data Heatmap

yellow is missing

green is not missing.

```
# plot missing data heatmap
cols = df.columns
colours = ['green', 'yellow']
sns.heatmap(df[cols].isnull(), cmap=sns.color_palette(colours))
```

```
<AxesSubplot:>
```

Missing Data Percentage List

```python
# show missing values percentage
for col in df.columns:
    pct_missing = np.mean(df[col].isnull())
    print('{}: {}%'.format(col, round(pct_missing*100)))
```

```
Date: 17%
No of Items: 75%
Price: 25%
Category: 8%
```

Missing Data Histogram

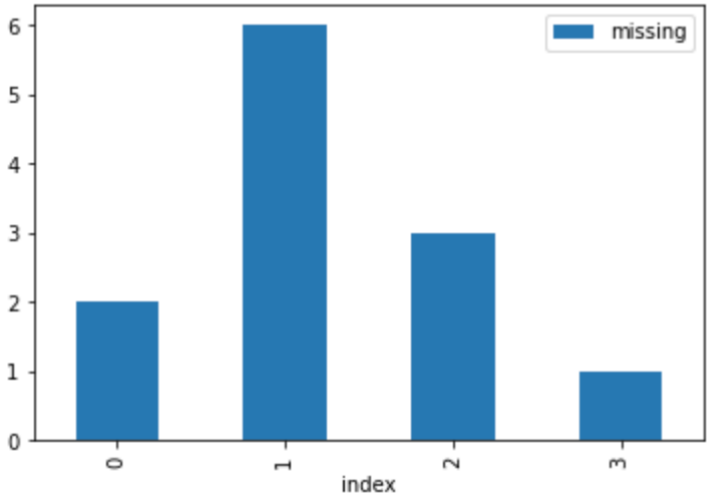
```
# plot missing values histogram
df_hist = df.copy()
for col in df_hist.columns:
    missing = df_hist[col].isnull()
    num_missing = np.sum(missing)

    if num_missing > 0:
        df_hist['{}_ismissing'.format(col)] = missing

ismissing_cols = [col for col in df_hist.columns if 'ismissing' in col]
df_hist['missing'] = df_hist[ismissing_cols].sum(axis=1)

df_hist['missing'].value_counts().reset_index().sort_values(by='index').plot.bar(x='index', y='missing')
```

```
<AxesSubplot:xlabel='index'>
```



Drop the observation

Drop rows with a lot of missing values

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Drop a feature

Drop columns with a lot of missing values

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7	25.11.2021	15.0	Black S
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Impute missing with median

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Impute missing with the most frequent element

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9	27.11.2021	NaN	15.0	Black S
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11	29.11.2021	NaN	10.0	Hoodie

Replace the missing with *a* specific value (code)

Missing

-999

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